

Document 18: Research & Final Year Project Guidelines

Institution: Hyderabad Institute of Technology (HIT)

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1. Academic Scope and Mandatory Project Execution

The Major Final Year Project is the culmination of the four-year B.Tech degree program at the Hyderabad Institute of Technology. It is designed to test a student's ability to synthesize theoretical knowledge and apply it to a complex, real-world engineering problem.

As outlined in the university's curricular framework, the Major Project is executed during the eighth and final semester. It carries a massive academic weight of exactly sixteen credits.

Crucially, the Major Project and the Semester-Long Industry Internship are mutually exclusive academic tracks. If a student successfully secures a six-month corporate internship and receives an official No-Objection Certificate (NOC) as per the Internship Policy, they are completely exempted from the Major Project. Conversely, any student who does not secure a corporate internship, or who fails to meet the 7.0 CGPA and zero-backlog eligibility criteria for internships, is legally mandated to remain on campus and execute the Major Project under the direct supervision of an HIT faculty member.

2. Team Formation Rules and Supervisor Allocation

Engineering is inherently collaborative; therefore, the university enforces strict regulations regarding team composition to simulate corporate project dynamics.

A Major Project team must consist of a minimum of three and a maximum of four students. Solo projects or two-person teams are strictly prohibited by the Academic Senate, as they fail to demonstrate teamwork and task delegation. The only exception to the solo-project ban is if a student has independently filed a provisional patent for a working hardware prototype prior to the start of the seventh semester, in which case they may apply to the Dean of Research for a "Solo Execution Waiver."

Teams must be formed within the same parent department (e.g., four CSE students). Cross-departmental multidisciplinary teams (e.g., two Mechanical students and two Electronics students building a drone) are highly encouraged but require a specialized "Joint-Supervision" approval from the Heads of both respective departments.

Once a team is formed, they must submit a Project Domain Preference form. The Departmental Project Review Committee (PRC) utilizes a lottery-based algorithm to assign an internal Faculty Supervisor to each team. A single faculty member is legally capped at supervising a maximum of three project teams per academic year to ensure they can dedicate sufficient mentorship time to each group. Students are not permitted to change their assigned supervisor once the semester officially begins.

3. The Four-Phase Evaluation and Milestone Tracking

The sixteen credits of the Major Project are not evaluated in a single final exam. The evaluation is distributed across a rigorous, four-phase milestone tracking system spread over the entire eighth semester. The total evaluation is marked out of 200 points.

1. **Phase 0: The Synopsis Pitch (Week 2):** The team must present a three-page synopsis outlining their problem statement, proposed methodology, and hardware/software requirements to the PRC. This is a qualifying round. If the PRC deems the project too simplistic (e.g., a basic library management system) or overly ambitious without clear feasibility, the pitch is rejected. The team has exactly one week to pitch a completely new idea, or they will be marked as "Failed" for the entire semester.
2. **Phase 1: Literature Review and Architecture Design (Week 6 - 40 Marks):** The team must present a comprehensive review of at least fifteen existing IEEE research papers related to their domain and showcase their system architecture diagrams (UML, Circuit Schematics, or Data Flow Diagrams).
3. **Phase 2: Partial Implementation and Testing (Week 11 - 60 Marks):** The team must demonstrate a working prototype of at least 50% of their proposed system. For software projects, the core backend logic and database must be fully functional. For hardware projects, the primary sensor arrays or mechanical chassis must be assembled and tested.
4. **Phase 3: Final Defense and Viva Voce (Week 16 - 100 Marks):** The team presents the fully completed project to an External Examiner invited from a Tier-1 institution (like IIT or NIT). The panel will heavily scrutinize the project's real-world viability, execution, and the individual technical contribution of each team member.

4. Plagiarism Thresholds and AI-Generation Penalties

Academic integrity in research is non-negotiable. Before a team is allowed to submit their final thesis report, the document must be processed through the university's official Turnitin plagiarism detection software.

The university enforces strict mathematical thresholds for textual similarity:

- **Maximum Overall Similarity:** The total overall similarity index of the thesis report must strictly be equal to or less than 15%.
- **Single-Source Limitation:** No more than 5% of the text can be matched to any single external source.
- **Penalty Bracket (16% to 25%):** If the Turnitin report shows an overall similarity between 16% and 25%, the thesis is not instantly rejected, but the team suffers a massive, unappealable deduction of 20 marks from their Final Defense score.
- **Rejection Bracket (Above 25%):** If the similarity index exceeds 25%, the thesis is immediately classified as plagiarized. The project is canceled, the team is awarded an automatic F-grade, and all members suffer an academic Year-Back.

Furthermore, utilizing Generative AI (such as ChatGPT, GitHub Copilot, or Claude) to write entire chapters of the thesis report or generate core algorithms without explicit citation is strictly prohibited. If the PRC discovers undocumented, AI-generated content during the code review, the team will face the Disciplinary Action Committee for academic malpractice.

5. Hardware Procurement and Institutional Property Rights

Students are expected to bear the financial cost of their project materials. The university does not provide direct funding for standard B.Tech major projects.

If a team is building a hardware-centric project (e.g., an autonomous rover, a smart grid inverter, or a customized CNC machine), they must purchase the microcontrollers, motors, and sensors out of their own pockets. However, if the project is deemed exceptionally innovative, the team can apply for the "Vice-Chancellor's Hardware Reimbursement Grant." If approved, the university will reimburse up to ₹15,000 against valid GST invoices.

Regardless of whether the students paid for the parts or the university reimbursed them, all completed physical hardware projects must be surrendered to the departmental laboratory after the Final Defense. Students are not permitted to take their hardware prototypes home after graduation. The university retains the physical models for junior demonstrations and NAAC accreditation showcases.

6. Thesis Formatting, Color Codes, and Final Submission

The Final Project Thesis is an official academic document that is permanently archived in the university library. Therefore, it must adhere to extremely rigid formatting and binding guidelines.

The thesis must be printed on A4-sized executive bond paper. It must have a minimum length of 80 pages and a maximum length of 150 pages, excluding the appendix and source code blocks. The font must strictly be Times New Roman, size 12, with 1.5 line spacing.

The final submission requires three physical hardbound copies. The exterior hardcover must be bound in a specific fabric color dictated by the student's parent department to ensure easy cataloging in the central library:

- **Computer Science & Engineering (CSE) and AI & DS:** Navy Blue fabric with Gold embossed lettering.
- **Electronics & Communication Engineering (ECE):** Maroon fabric with Gold embossed lettering.
- **Mechanical Engineering (MEC):** Black fabric with Silver embossed lettering.
- **Civil Engineering (CIV):** Dark Green fabric with Silver embossed lettering.

If a team submits a thesis bound in the incorrect color, or if the page length falls below 80 pages, the library will reject the submission, and the students will not receive their final No-Dues clearance.

7. Research Publication Mandates and Bonus Credits

HIT aggressively pushes its students to contribute to the global academic community. While publishing a research paper is not strictly mandatory to pass the B.Tech degree, the university offers massive academic incentives for teams that successfully publish their Major Project findings.

If a project team manages to get their research paper accepted and published in a Scopus-indexed journal or an official IEEE International Conference before the date of their Final Defense, they are granted the **"Research Excellence Bonus."**

- This bonus instantly adds 15 bonus marks to their Final Defense score (capping at the maximum of 100).
- Furthermore, if the publication requires a registration fee (often upwards of ₹8,000 for IEEE conferences), the university's Research & Development Cell will reimburse 100% of the publication fee, provided the university's name is officially listed as the primary institutional affiliation on the published paper.
- Papers published in unverified, "predatory," or non-indexed open-access journals will not be recognized, and no bonus marks or financial reimbursements will be awarded.